CHAPTER 3 Microbiology

SYLLABUS

Viruses— Structure and Classification; Microbial Classification and Diversity (Bacterial, Algal and Fungal); Methods in Microbiology; Microbial Growth and Nutrition; Aerobic and Anaerobic Respiration; Nitrogen Fixation; Microbial Diseases and Host-Pathogen Interaction.

CHAPTER ANALYSIS

Topic	GATE 2010	GATE 2011	GATE 2012	GATE 2013	GATE 2014	GATE 2015	GATE 2016	GATE 2017	GATE 2018	GATE 2019
Viruses: Structure and Classification		1					1	1		1
Microbial Classification and Diversity: Bacterial, Algal and Fungal	1		1	2	1		1	1		
Methods in Microbiology								1		
Microbial Growth and Nutrition	2							1		
Aerobic and Anaerobic Respiration					1				2	
Nitrogen Fixation						1	1		1	
Microbial Diseases and Host-Pathogen Interaction	2	2		1		4	1	2	1	2
Applications of Microbiology		1				1			1	

QUESTIONS

1. The typical coenzyme present in the methanogens is
   (A) coenzyme A. (B) coenzyme Q.
   (C) coenzyme M. (D) None of the above.
   (GATE 2002; 1 Mark)

2. Under stress conditions bacteria accumulate
   (A) ppGpp (Goanosine tetraphosphate).
   (B) pppGpp (Guanosine pentaphosphate).
   (C) both ppGpp and pppGpp.
   (D) either ppGpp or pppGpp.
   (GATE 2010; 1 Mark)

3. Which one of the following does not belong to the domain of Bacteria?
   (A) Cyanobacteria (B) Proteobacteria
   (C) Bacteroids (D) Methanobacterium
   (GATE 2010; 1 Mark)

4. A culture of bacteria is infected with bacteriophage at a multiplicity of 0.3. The probability of a single cell infected with 3 phages is
   (A) 0.9 (B) 0.27
   (C) 0.009 (D) 0.027
   (GATE 2010; 1 Mark)

5. Match the following antibiotics in Group I with their mode of action in Group II.

Group I	Group II
P. Chloramphenicol	1. Binds to DNA gyrase
Q. Norfloxacin	2. Binds to RNA Polymerase
R. Puromycin	3. Inhibits peptidyl transferase
S. Rifampicin	4. Mimics aminoacyl-tRNA

   (A) P–1; Q–3; R–2; S–4 (B) P–3; Q–1; R–2; S–4
   (C) P–3; Q–1; R–4; S–2 (D) P–4; Q–2; R–3; S–1
   (GATE 2010; 2 Marks)

6. Match Group I with Group II.

Group I	Group II
P. *Staphylococcus aureus*	1. Biofilms
Q. *Candida albicans*	2. Bacteriocins
R. *Mycobacterium tuberculosis*	3. Methicillin resistance
S. *Lactobacillus lactis*	4. Isoniazid

(A) P–1; Q–4; R–2; S–3
(B) P–2; Q–3; R–1; S–4
(C) P–3; Q–1; R–4; S–2
(D) P–1; Q–2; R–4; S–3

(GATE 2010; 2 Marks)

7. Diphtheria toxin, tetracycline and streptomycin inhibit
(A) DNA repair.
(B) DNA replication.
(C) transcription.
(D) translation.

(GATE 2011; 1 Mark)

8. Match the viruses in Group I with their host cell receptors in Group II.

Group I	Group II
P. Hepatitis A virus	1. Heparan sulphate
Q. Human immunodeficiency virus	2. Acetylcholine receptor
R. Rabies virus	3. CD4 protein
S. Herpes simplex virus type I	4. Alpha-2 macroglobulin

(A) P–1; Q–3; R–2; S–4
(B) P–3; Q–4; R–1; S–2
(C) P–4; Q–3; R–2; S–1
(D) P–2; Q–3; R–1; S–4

(GATE 2011; 2 Marks)

9. Match the products in Group I with their respective organisms in Group II.

Group I	Group II
P. Glycerol	1. *Corynebacterium glutamicum*
Q. Glutamic acid	2. *Alcaligenes faecalis*
R. Curdlan	3. *Dunaliella salina*
S. Amphotericin B	4. *Streptomyces nodosus*

(A) P–2; Q–1; R–3; S–4
(B) P–4; Q–2; R–1; S–3
(C) P–3; Q–1; R–2; S–4
(D) P–2; Q–1; R–4; S–3

(GATE 2011; 2 Marks)

10. Match items in Group I with Group II.

Group I	Group II
P. Alzheimer's disease	1. H1N1
Q. Mad cow disease	2. Hemoglobin
R. Sickle cell anemia	3. Prions
S. Swine flu	4. Amyloid

(A) P–4; Q–3; R–2; S–1
(B) P–3; Q–4; R–1; S–2
(C) P–2; Q–1; R–4; S–3
(D) P–1; Q–2; R–3; S–4

(GATE 2011; 2 Marks)

11. Match the organisms in Group I with the entries in Group II

Group I	Group II
P. *Clostridium*	1. Rods with teichoic acid in the cell wall
Q. *Escherichia*	2. Rods with endospores
R. *Vibrio*	3. Helical rods with flagella
S. *Bacillus*	4. Rods with LPS in the outer membrane
	5. Curved rods with polar flagella

(A) P–2; Q–4; R–5; S–1
(B) P–2; Q–1; R–5; S–4
(C) P–5; Q–4; R–2; S–3
(D) P–3; Q–2; R–1; S–4

(GATE 2012; 2 Marks)

12. Phylum proteobacteria is subdivided into α, β, γ, δ and ε-proteobacteria based on
(A) G + C content
(B) 23S rRNA sequences
(C) tRNA sequences
(D) 16S rRNA sequences

(GATE 2013; 1 Mark)

13. Match the cell structures in Group I with the organisms in Group II.

Group I	Group II
P. Endospores	1. *Methanobacterium*
Q. Bipolar flagella	2. *Treponema*
R. Pseudomurine in cell wall	3. *Spirillum*
S. Periplasmic flagella	4. *Clostridium*

(A) P–4; Q–3; R–1; S–2
(B) P–4; Q–3; R–2; S–1
(C) P–3; Q–4; R–1; S–2
(D) P–4; Q–1; R–3; S–2

(GATE 2013; 2 Marks)

14. Match the antibiotics in Group I with the targets in Group II.

Group I	Group II
P. Sulfonamide	1. Peptidoglycan synthesis
Q. Quinolones	2. Peptide chain elongation
R. Erythromycin	3. Folic acid biosynthesis
S. Cephalosporin	4. Topoisomerase

(A) P–3; Q–4; R–1; S–2 **(B)** P–2; Q–4; R–3; S–1
(C) P–4; Q–1; R–2; S–3 **(D)** P–3; Q–4; R–2; S–1

(GATE 2013; 2 Marks)

15. A bacterium belonging to cocci group has a diameter of 2μ m. The numerical value of the ratio of its surface area to volume (in $\mu\ m^{-1}$) is _____.

(GATE 2014; 1 Mark)

16. The most plausible explanation for a sudden increase of the respiratory quotient (RQ) of a microbial culture is that

(A) cells are dying.
(B) yield of biomass is increasing.
(C) the fermentation rate is increasing relative to respiration rate.
(D) the maintenance rate is decreasing.

(GATE 2014; 1 Mark)

17. Which one of the following organisms is used for the determination of phenol coefficient of a disinfectant?

(A) *Salmonella typhi* **(B)** *Escherichia coli*
(C) *Candida albicans* **(D)** *Bacillus psychrophilus*

(GATE 2015; 1 Mark)

18. Which one of the following is not a product of denitrification in *Pseudomonas*?

(A) N_2 **(B)** N_2O
(C) NO_2 **(D)** NH_4^+

(GATE 2015; 1 Mark)

19. Match the compounds in Group I with the correct entries in Group II.

Group I	Group II
P. Cyanide	1. K^+ ionophore
Q. Antimycin A	2. Electron transfer from cytochrome *b* to cytochrome c_1
R. Valinomycin	3. F_1 subunit of ATP synthase
S. Aurovertin	4. Cytochrome oxidase
	5. Adenine nucleotide translocase

(A) P–5; Q–2; R–3; S–1 **(B)** P–5; Q–2; R–1; S–3
(C) P–4; Q–2; R–1; S–3 **(D)** P–4; Q–5; R–3; S–1

(GATE 2015; 2 Marks)

20. Match the drugs in Group I with their mechanism of action in Group II.

Group I	Group II
P. Paclitaxel	1. Inhibits protein translation
Q. Colchicine	2. Inhibits microtubule depolymerization
R. Etoposide	3. Inhibits DNA replication
S. Methotrexate	4. Alkylates DNA
	5. Inhibits dihydrofolate reductase
	6. Inhibits microtubule polymerization

(A) P–1; Q–6; R–3; S–4
(B) P–2; Q–6; R–3; S–5
(C) P–1; Q–3; R–6; S–5
(D) P–2; Q–3; R–6; S–4

(GATE 2015; 2 Marks)

21. Match the microorganisms in Group I with their fermentation products in Group II.

Group I	Group II
P. *Leuconostoc mesenteroides*	1. Cobalamin
Q. *Rhizopus oryzae*	2. Sorbose
R. *Gluconobacter suboxydans*	3. Dextran
S. *Streptomyces olivaceus*	4. Lactic acid
	5. Butanol

(A) P–5; Q–4; R–2; S–1
(B) P–5; Q–3; R–2; S–4
(C) P–3; Q–4; R–1; S–2
(D) P–3; Q–4; R–2; S–1

(GATE 2015; 2 Marks)

22. Match the antibiotics in Group I with their modes of action in Group II.

Group I	Group II
P. Chloramphenicol	1. Inhibits protein synthesis by acting on 30S ribosomal subunit
Q. Rifampicin	2. Interferes with DNA replication by inhibiting DNA gyrase
R. Tetracycline	3. Inhibits protein synthesis by acting on 50S ribosomal subunit
S. Quinolone	4. Interferes with RNA polymerase activity
	5. Inhibits P-lactamase activity

(A) P–1; Q–2; R–3; S–5
(B) P–3; Q–4; R–1; S–2
(C) P–3; Q–2; R–1; S–4
(D) P–1; Q–4; R–3; S–2

(GATE 2015; 2 Marks)

23. Bacteria with two or more flagella at one or both ends are called

(A) amphitrichous. **(B)** peritrichous.
(C) lophotrichous. **(D)** atrichous.

(GATE 2016; 1 Mark)

24. Which family of viruses has single stranded DNA?

(A) Herpesviridae (B) Poxviridae
(C) Retroviridae (D) Parvoviridae

(GATE 2016; 1 Mark)

25. Which of the following are true for *Treponema pallidum*?

P. It is the causative agent of syphilis
Q. It is a spirochete
R. It is a non-motile bacterium
S. It is generally susceptible to penicillin

Choose the correct combination.

(A) P, Q and R only (B) P, Q and S only
(C) P, R and S only (D) Q, R and S only

(GATE 2016; 1 Mark)

26. Which combination of the following statements is correct for cyanobacteria?

P. They can perform oxygenic photosynthesis.
Q. Usually filamentous forms are involved in nitrogen fixation.
R. Nitrogen fixation occurs in heterocyst.
S. They cannot grow in a mineral medium exposed to light and air.

(A) P, Q and R (B) P, S and R
(C) Q, R and S (D) P, Q and S

(GATE 2016; 2 Marks)

27. Which one of the following organisms is an indicator of fecal contamination?

(A) *Escherichia coli*
(B) *Streptococcus lactis*
(C) *Bacillus subtilis*
(D) *Lactobacillus acidophilus*

(GATE 2017; 1 Mark)

28. Determine the correctness or otherwise of the following Assertion (*A*) and Reason (*R*).

Assertion: Gram negative bacteria show staining with safranin.

Reason: Gram negative bacteria have an outer membrane with lipopolysaccharides.

(A) Both (*A*) and (*R*) are true and (*R*) is the correct reason for (*A*).
(B) Both (*A*) and (*R*) are true but (*R*) is not the correct reason for (*A*).
(C) Both (*A*) and (*R*) are false.
(D) (*A*) is true but (*R*) is false.

(GATE 2017; 1 Mark)

29. Match the organisms in Group I with the characteristics in Group II.

Group I	Group II
P. *Methanococcus*	1. Halophile
Q. *Dunaliella*	2. Acidophile
R. *Sulfolobus*	3. Mesophile
S. *Escherichia*	4. Barophile

(A) P–4; Q–3; R–2; S–1 (B) P–3; Q–2; R–4; S–1
(C) P–2; Q–1; R–4; S–3 (D) P–4; Q–1; R–2; S–3

(GATE 2017; 2 Marks)

30. Which one of the following organisms is responsible for crown gall disease in plants?

(A) *Xanthomonas campestris*
(B) *Rhizobium etli*
(C) *Agrobacterium tumefaciens*
(D) *Erwinia stewartii*

(GATE 2017; 2 Marks)

31. For an *E. coli* culture in the exponential phase of growth, optical density (OD) at 540 nm is 0.3 at 2 hours and 0.6 at 4 hours. Assuming that the measured OD is linearly proportional to the number of *E. coli* cells, the growth rate (per hour) for this culture is_____.

(GATE 2017; 2 Marks)

32. One hundred *E. coli* cells are each infected by a single λ phage particle. The ratio of the number of phage particles committing to lysogeny to those committing to lysis is 4:1. Assuming that the average burst size is 80, the number of free phage particles released after one round of infection is _____.

(GATE 2017; 2 Marks)

33. Which one of the following is an example of a neurotoxin?

(A) Cholera toxin (B) Streptolysin-O
(C) Botulinum toxin (D) Diphtheria toxin

(GATE 2018; 1 Mark)

34. In leguminous plants, both the rhizobium genes and the plant genes influence nodulation and nitrogen fixation. Which one of the following functions is not encoded by the host plant genes?

(A) Production of inducers that modify rhizobial cell wall.
(B) Production of flavonoid inducers.
(C) Establishment of contact between bacteria and legume.
(D) Root hair curling.

(GATE 2018; 2 Marks)

35. Which of the following are true about bacterial superoxide dismutase?

P. Present in obligate aerobes.
Q. Present in facultative anaerobes.

R. Present in aerotolerant anaerobes.
S. Absent in obligate aerobes.

(A) P and Q only
(B) P, Q and R only
(C) P and R only
(D) Q and S only

(GATE 2018; 2 Marks)

36. Which of the following are true with regard to anaerobic respiration in bacteria?

P. The final electron acceptor is an inorganic substance other than molecular oxygen.
Q. The number of ATP molecules produced per glucose molecule is more than that produced in aerobic respiration.
R. The number of ATP molecules produced per glucose molecule is less than that produced in aerobic respiration.
S. Only substrate level phosphorylation is used to generate ATP.

(A) P and S only
(B) Q and S only
(C) P and R only
(D) P, Q and S only

(GATE 2018; 2 Marks)

37. Match the industrial products mentioned in Group I with their producer organisms in Group II

Group I	Group II
P. Citric acid	1. *Trichoderma viride*
Q. Cellulase	2. *Clostridium acetobutylicum*
R. Vitamin B_{12}	3. *Aspergillus niger*
S. Butanol	4. *Propionibacterium freudenreichii*

(A) P–4; Q–3; R–1; S–2
(B) P–3; Q–1; R–2; S–4
(C) P–2; Q–1; R–4; S–3
(D) P–3; Q–1; R–4; S–2

(GATE 2018; 2 Marks)

38. Which one of the following enzymes is encoded by human immunodeficiency virus (HIV) genome?

(A) Reverse transcriptase
(B) Phospholipase
(C) Phosphatase
(D) ATP synthase

(GATE 2019; 1 Mark)

39. Match the human diseases in Group I with the causative agents in Group II.

Group I	Group II
P. Amoebiasis	1. *Leishmania donovani*
Q. African sleeping sickness	2. *Trypanosoma cruzi*
R. Kala azar	3. *Entamoeba histolytica*
S. Chagas' disease	4. *Trypanosoma gambiense*

(A) P–3; Q–4; R–2; S–1
(B) P–3; Q–2; R–1; S–4
(C) P–3; Q–4; R–1; S–2
(D) P–4; Q–3; R–1; S–2

(GATE 2019; 1 Mark)

40. Tetracycline inhibits the

(A) interaction between tRNA and mRNA.
(B) translocation of mRNA through ribosome.
(C) peptidyl transferase activity.
(D) binding of amino-acyl tRNA to ribosome.

(GATE 2019; 1 Mark)

ANSWER KEY

1. (C)	2. (B)	3. (D)	4. (D)	5. (C)	6. (C)	7. (D)	8. (C)	9. (C)	10. (A)
11. (A)	12. (D)	13. (A)	14. (D)	15. (3 μm^{-1})	16. (C)	17. (A)	18. (D)	19. (C)	20. (B)
21. (D)	22. (B)	23. (C)	24. (D)	25. (B)	26. (A)	27. (A)	28. (A)	29. (D)	30. (C)
31. (0.346)	32. (1600)	33. (C)	34. (D)	35. (B)	36. (C)	37. (D)	38. (A)	39. (C)	40. (D)

ANSWERS WITH EXPLANATIONS

1. Topic: Microbial Classification and Diversity

Coenzyme M is a coenzyme required for methyl-transfer reactions in the metabolism of methanogens. The coenzyme is an anion with the formula $HSCH_2CH_2SO_3^-$. It is named 2-mercaptoethanesulfonate and abbreviated HS–CoM. The coenzyme is the C1 donor in methanogenesis.

Answer (C)

2. Topic: Microbial Growth and Nutrition

Bacteria use an array of sensory systems to monitor their environment to enable adaptation to stressful conditions. Most of these systems convert external stimuli into changes in the intracellular concentration of a secondary messenger molecule, which functions as a pleiotropic regulator of key molecular targets. There are three common

nucleotide-based secondary messengers in bacteria: cAMP, c-di-GMP and (p)ppGpp. (p)ppGpp is a collective term that refers to two related alarmone nucleotides, ppGpp and pppGpp, which are formed by the addition of a pyrophosphate moiety to the 3′ position of GDP and GTP respectively. When nutrient limitation slows down the growth rate, and cellular resources and energy are driven from ribosome biosynthesis towards the upkeep of general metabolism. Regulation of general metabolism by (p)ppGpp is exerted by its action on multiple processes: first, at the level of transcription, (p)ppGpp controls the expression of genes involved in amino acid biosynthesis; second, it regulates nucleotide metabolism by binding directly to the enzymes involved in nucleotide biosynthesis and uptake.

Answer (B)

3. **Topic: Microbial Classification and Diversity (Bacteria)**

Kingdom Monera has been divided into 2 domains: the domain Bacteria and the domain Archaea. Cyanobacteria, Proteobacteria and Bacteroides all fall under the domain of Bacteria. Methanobacterium belongs to the domain of Archaebacteria. These bacteria produce methane, or marsh gas, by reducing carbon-containing compounds to the gas methane. Hence, they are called methanogens.

Answer (D)

4. **Topic: Microbial Growth and Nutrition**

Multiplicity of infection (MOI) refers to the number of virions that are added per cell during infection.

Probability of single cell infected with x phages,

$$P(x) = x^m$$

where,

m = multiplicity of infection

Probability of infection with 3 phages,

$$P(3) = 3^{0.3} = 0.027$$

Answer (D)

5. **Topic: Antibiotics (Microbial Diseases)**

- Chloramphenicol is a bacteriostatic that inhibits protein synthesis. It prevents protein chain elongation by inhibiting the peptidyl transferase activity of the bacterial ribosome.
- Norfloxacin is a quinolone/fluoroquinolone antibiotic. Norfloxacin is bactericidal and its mode of action depends on blocking of bacterial DNA replication by binding itself to an enzyme called DNA gyrase, which allows the untwisting required to replicate one DNA double helix into two.
- Puromycin is a structural analogue of the 3′ end of aminoacyl-transfer RNA but differs from tRNA as the aminoacyl residue is linked to the ribose via an amide bond rather than an ester bond. Puromycin, like aminoacyl-tRNA, binds to the A site of the ribosome peptidyl-transferase center. When the A site is occupied by puromycin, peptidyl-transferase links the peptide residues of the peptidyl-tRNA in the ribosomal P site covalently to puromycin. Since the amide bond cannot be cleaved by the ribosome, no further peptidyl transfer takes place, and the peptidyl-puromycin complex falls off the ribosome.
- Rifampin acts via the inhibition of DNA-dependent RNA polymerase, leading to a suppression of RNA synthesis and cell death.

Answer (C)

6. **Topic: Microbial Diseases**

- Methicillin-resistant *Staphylococcus aureus* (MRSA) is a bacterium that causes infections in different parts of the body. It is tougher to treat than most strains of *Staphylococcus aureus* because it's resistant to some commonly used antibiotics.
- A major virulence attribute of *Candida albicans* is its ability to form biofilms, densely packed communities of cells adhered to a surface. These biofilms are intrinsically resistant to conventional antifungal therapeutics, the host immune system, and other environmental factors, making biofilm-associated infections a significant clinical challenge.
- Isoniazid, also known as isonicotinylhydrazide (INH), is an antibiotic used for the treatment of tuberculosis against *Mycobacterium tuberculosis*. It may also be used for atypical types of mycobacteria, such as *M. avium*, *M. kansasii*, and *M. xenopi*.
- Bacteriocins are biologically active proteins or protein complexes displaying a bactericidal mode of action exclusively towards Gram-positive bacteria and particularly closely related species. Bacteriocins are produced by fifteen strains of *Lactococcus lactis*. In the last years, the bacteriocins of lactic acid bacteria have attracted much attention because of their potential to increase safety and to extend shelf life of food.

Answer (C)

7. **Topic: Microbial Diseases**

Tetracycline and streptomycin are the antibiotics involved in the inhibition of translation. Tetracycline blocks the binding of aminoacyl-tRNA, to the small subunit in order to inhibit translation. Diphtheria toxins acts on translocase (EF-2) and inhibits protein synthesis. Lastly, streptomycin attacks on small subunit and prevents the peptide bond formation.

Answer (D)

8. Topic: Viruses—Structure and Classification

- Alpha-2-Macroglobulin (α-2-M) is a large (720 kDa) plasma protein found in the blood. The formation of interferon and alpha 2-macroglobulin show active infection involvement in immune response during hepatitis A virus.
- HIV is a virus spread through certain body fluids that attacks the body's immune system, specifically the CD4 cells, often called T cells.
- The acetylcholine receptor acts as a cellular receptor for rabies virus.
- HSV-1 use the cell surface glycosaminoglycan heparan sulfate as a receptor.

Answer (C)

9. Topic: Application of Microbiology

- *Dunaliella salina* (halophile green micro-algae) is an important natural source for production of glycerol. It is produced by two different ways. First is photosynthetic fixation of carbon dioxide, and the second is the starch conversion.
- *Corynebacterium glutamicum* (Gram-positive) is a, rod-shaped bacteria that is used commercially for large-scale production of glutamic acid amino acids.
- *Alcaligenes faecalis* is a species of Gram-negative, rod-shaped bacteria used for production of curdlan, which is a water-insoluble linear Beta-1,3-glucan, a high-molecular-weight polymer of glucose.
- *Streptomyces nodosus* is amphotericin B (antifungal medication) producing bacterium.

Answer (C)

10. Topic: Microbial Diseases

- Alzheimer's disease (AD) is a progressive disease that destroys memory and other important mental functions. These are the extracellular plaque deposits of the β-amyloid peptide.
- Mad cow disease, or bovine spongiform encephalopathy (BSE), is a transmissible, degenerative, and fatal disease which affects the brain of adult cattle. It results from infection caused by prion.
- Sickle cell anemia (SCA) occurs because of an abnormality in the oxygen-carrying protein hemoglobin found in red blood cells.
- H1N1 flu (Swine flu) so called because in the past, the people who caught it had direct contact with pigs.

Answer (A)

11. Topic: Microbial Classification and Diversity (Bacteria)

- Physiological adaptation into the endospore production is a very important existence feature of some Gram-positive rods shaped bacteria, such as species of *Clostridium*.
- Lipopolysaccharides (LPS), also known as lipoglycans and endotoxins, are large molecules consisting of a lipid and a polysaccharide composed of O-antigen, they are found mainly in the plasma membrane of Gram-negative bacteria such as *E. coli*.
- *Vibrio* is a Gram-negative bacterium, having a curved-rod shape (comma shape), many species of which can cause foodborne diseases which are usually associated with eating undercooked seafood.
- The cell wall of the Gram-positive bacteria such as *Bacillus* has a polymer composed mainly of peptidoglycan (PG) and teichoic acids, both of which have important role in maintenance of the structural integrity and the shape of the cell.

Answer (A)

12. Topic: Microbial Classification and Diversity (Bacteria)

Phylum proteobacteria contains Gram-negative bacteria which can be sub-divided into 6 classes based 16S rRNA-based LTP released by the All-Species Living Tree Project.

Answer (D)

13. Topic: Microbial Classification and Diversity (Bacteria)

- Endospore formation occurs in extreme conditions, i.e., deficiency of nutrients. It usually occurs in Gram-positive bacteria, for example, *Bacillus* and *Clostridium*.
- Apart from rod and coccus shape, spiral bacteria (helical) shape, form the third major morphological category of bacteria. The two types of spiral cells are spirillum and spirochete. *Spirillum* bacteria have bipolar tufts of flagella.
- Pseudopeptidoglycan (also known as pseudomurein) is a major cell wall component of some Archaea (*Methanobacterium*) that differs from bacterial peptidoglycan in chemical structure. The basic components are *N*-acetylglucosamine (NAG) and *N*-acetyltalosaminuronic acid (peptidoglycan has *N*-acetylmuramic acid (NAM) instead), which are interlinked by β-1,3-glycosidic bonds.
- Spirochete periplasmic flagella (PFs), including those from Brachyspira (*Serpulina*), Spirochaeta, *Treponema*, and *Leptospira* species. A given periplasmic flagellum is attached sub-terminally at only one end of the cell cylinder, and it resides within the periplasmic space.

Answer (A)

14. Topic: Antibiotics (Microbial Diseases)

- Sulfonamides (antibiotics) hinder with folic acid production by preventing addition of para-aminobenzoic acid (PABA) into the folic acid molecule for the enzyme dihydropteroate synthetase.

- Quinolones (antibiotics) inhibit the enzymatic activities of two members of the topoisomerase class of enzymes–DNA gyrase and topoisomerase IV and stimulate the cleavage of DNA in these enzyme-DNA complexes.
- Erythromycin (antibiotics) inhibits protein formation by binding to the 23S rRNA (50S subunit of ribosome) of the bacteria and blocking the exit of the growing peptide chain.
- Cephalosporins and penicillins possess a similar mechanism of action: inhibition of bacterial cell wall peptidoglycan synthesis by inhibition of penicillin-sensitive enzymes.

Answer (D)

15. Topic: Microbial Classification and Diversity (Bacteria)

Cocci bacteria have sphere shape.

Surface area of sphere $= 4\pi r^2$

Volume of sphere $= \frac{4}{3}\pi r^3$

Ratio of surface area to volume $= \frac{\frac{4}{3}\pi r^3}{4\pi r^2} = \frac{3}{r}$

Given, diameter of cocci $= 2\mu m$

So, radius of cocci $= 1\mu m$

Ratio of surface area to volume $= \frac{3}{1\mu m} = 3\mu m^{-1}$

Answer ($3\ \mu m^{-1}$)

16. Topic: Aerobic and Anaerobic Respiration

Respiratory quotient (RQ) represents the ratio of the volume of carbon dioxide produced to the volume of oxygen consumed in respiration. Anaerobic respiration that takes place in microorganisms is termed as fermentation. It involves a controlled step-wise cellular degradation or incomplete oxidation of organic compounds in which energy is made available to cells in absence of oxygen and carbon dioxide is evolved. If the rate of fermentation increases with respect to respiration rate, more amount of carbon dioxide will be evolved, thus increasing the respiratory quotient.

Answer (C)

17. Topic: Sterilization and Disinfectant (Microbial Diseases)

Phenol coefficient of a disinfectant is calculated by dividing the dilution of test disinfectant by the dilution of phenol that disinfects under predetermined conditions.

Salmonella typhi bacterial suspension is used to evaluate the bactericidal activity of test compound. Coefficient is calculated by dividing the dilution of test disinfectant by dilution of phenol in same conditions.

Answer (A)

18. Topic: Nitrogen Fixation

Denitrification is the process in which nitrates are reduced to gaseous nitrogen dioxide, nitrogen oxide, nitrogen monoxide or elemental nitrogen. The bacteria involved in denitrification are usually aerobic and under anaerobic conditions they use nitrate instead of oxygen as a hydrogen acceptor. These include *Pseudomonas* species. They breakdown NO_3^- to metabolize oxygen and nitrogen gas is released.

Answer (D)

19. Topic: Antibiotics (Microbial Diseases)

- Cyanide is a chemical compound that contains monovalent combining group CN. This group consists of a carbon atom triple- bonded to a nitrogen atom. It is considered to be toxic as it binds to cytochrome *c* oxidase, i.e., the fourth complex in the electron transport chain. It attaches to the iron within this protein complex and inhibits the normal activity of the complex system. It binds tightly so that it cannot transport any electrons to oxygen. This blocks the further passage of electrons throughout the chain, halting ATP production.
- Antimycin A is an inhibitor of cellular respiration, specifically oxidative phosphorylation. Antimycin A binds to the Qi site of cytochrome *c* reductase, inhibiting the oxidation of ubiquinone in the Qi site of ubiquinol thereby disrupting the Q-cycle of enzyme turn over. Cytochrome *c* reductase is a central enzyme in the electron transport chain of oxidative phosphorylation. The inhibition of this reaction disrupts the formation of the proton gradient across the inner membrane of the mitochondria. The production of ATP is subsequently inhibited, as protons are unable to flow through the ATP synthase complex in the absence of a proton gradient.
- Valinomycin functions as a potassium-specific transporter and facilitates the movement of potassium ions through lipid membranes down the electrochemical potential gradient. The stability constant for the potassium-valinomycin complex is nearly 100,000 times larger than that of the sodium-valinomycin complex. This difference is important for maintaining the selectivity of valinomycin for the transport of potassium ions (and not sodium ions) in biological systems.
- The mitochondrial F_1F_0 ATPase performs the terminal step of oxidative phosphorylation. Aurovertin is an antibiotic that binds to the beta subunits in the F_1 domain and inhibits F_1F_0 ATPase catalyzed ATP synthesis in preference to ATP hydrolysis.

Answer (C)

20. Topic: Antibiotics (Microbial Diseases)

- Paclitaxel is one of several cytoskeletal drugs that target tubulin. It stabilizes the microtubule polymer and protects it from disassembly. Chromosomes thus are unable to achieve a metaphase spindle configuration. This blocks the progression of mitosis and prolonged activation of the mitotic checkpoint triggers apoptosis or reversion to the G_0-phase of the cell cycle without cell division.
- Colchicine inhibits microtubule polymerization by binding to tubulin, one of the main constituents of microtubules. As the availability of tubulin is essential to mitosis, colchicine may inhibit mitosis.
- Etoposide forms a ternary complex with DNA and the topoisomerase II enzyme (which aids in DNA unwinding), thus prevents re-ligation of the DNA strands, and by doing so causes DNA strands to break.
- Methotrexate is an antimetabolite of the antifolate type. For cancer, methotrexate competitively inhibits dihydrofolate reductase (DHFR), an enzyme that participates in the tetrahydrofolate synthesis. The affinity of methotrexate for DHFR is about 1000-fold that of folate. DHFR catalyzes the conversion of dihydrofolate to the active tetrahydrofolate.

Answer (B)

21. Topic: Application of Microbiology

- *Leuconostoc mesenteroides* is a bacterial species sometimes associated with fermentation, under conditions of salinity and low temperatures. When grown in sucrose solution, it converts the sugar to dextrans having mostly alpha 1,6 linkages, but 1,2, 1,3, and 1,4 linkages are also present.
- *Rhizopus oryzae* is capable of producing high levels of lactic acid by the fermentation of glucose. Yields typically vary over 60–80%, with the remaining glucose diverted primarily into ethanol fermentation.
- Resting cell of *Gluconobacter suboxydans* is used in the biotransformation of D-sorbitol to L-sorbose, a key step in the industrial manufacture of Vitamin C (ascorbic acid).
- Cobalmin, commonly known as vitamin B_{12} is obtained through microbial fermentation by *Streptomyces olivaceus*. It is grown in nutrient medium containing glucose as carbon source. Cobalt chloride is added as precursor at the beginning during the production.

Answer (D)

22. Topic: Antibiotics (Microbial Diseases)

- Chloramphenicol is a broad-spectrum antibiotic derived from *Streptomyces venequelae* with primarily bacteriostatic activity. It does so by diffusing through the bacterial cell wall and reversibly binding to the bacterial 50S ribosomal subunit. It halts the production of proteins in the cell.
- Rifampicin, also known as rifampin is a semisynthetic antibiotic produced by *Streptomyces mediterranei*. It blocks RNA transcription by inhibiting bacterial DNA-dependent RNA polymerase.
- Tetracyclines are obtained from species of *Streptomyces*. They have the widest spectrum of activity and are effective against both Gram-positive and Gram-negative bacteria. Inhibits protein synthesis by acting on 30S ribosomal subunit. They inhibit protein synthesis by blocking the attachment of charged aminoacyl-tRNA to the A site on the ribosome. They bind to the 30S subunit of microbial ribosomes, thus, prevent introduction of new amino acids to the nascent peptide chain.
- Quinolones is a group of synthetic bactericidal analogs of nalidixic acid. They are effective against many Gram-positive and Gram-negative bacteria. They inhibit DNA synthesis by blocking DNA gyrase, the enzyme that unwinds the DNA double helix preparatory to its replication.

Answer (B)

23. Topic: Microbial Classification and Diversity (Bacteria)

Arrangements of bacterial flagella:

- Polar, monotrichous (single flagellum at one end).
- Polar, amphitrichous (flagella at both the ends).
- Lophotrichous (with tuft of flagella at one or both ends).
- Amphicephatotrichous or cephalotrichous (with many flagella at one end).
- Peritrichous (flagella distributed all over).
- Artichous (no flagellum present).

Answer (C)

24. Topic: Viruses—Structure and Classification

- Parvoviridae are non-enveloped, measuring 18–26 nm in diameter and represent some of the smallest known vertebrate viruses. They have icosahedral symmetry and a genome consisting of a single strand of DNA.
- Herpesviridae family has double stranded (ds) DNA.
- Poxviridae has ds DNA.
- Retroviridae has ss RNA.

Answer (D)

25. Topic: Microbial Diseases

Treponema pallidum is an active motile spirochete, it is not stained with gram staining; but stained by silver

impregnation method or with fluorescent dye; and is visualized under dark-ground microscopy. it causes syphilis in adults and congenital syphilis in newborn babies who acquire transplacental infection. Syphilis is usually treated with benzathine penicillin G. There is no vaccine available and the recovery does not confer immunity. Congenital syphilis occurs in babies when *T. pallidum* cross the placenta from mother to baby. Syphilis is a sexually-transmitted diseases.

Answer (B)

26. Topic: Nitrogen Fixation

Cyanobacteria are photosynthetic prokaryotes that also help in nitrogen fixation. The genetic material in cyanobacteria is not contained within a membrane-limited organelle. They have specialized nitrogen-fixing cells called heterocyst. These cells have thickened, multilayered cell walls that restrict the diffusion of oxygen. Also, they have high respiratory activity to maintain low intracellular oxygen concentration.

Answer (A)

27. Topic: Microbial Diseases and Host Pathogen Interaction

Number and type of bacteria vary with specific urinary tract infections. One of the most common is *Escherichia coli* of human colon. They are found to consume undigested food and complete its life cycle in their hosts. It causes urinary tract infections, "traveler's diarrhea," nosocomial infections.

E. coli serves as an indicator organism as it is always present in water contaminated with fecal material. Generally, it is more numerous than other organism and can be easily isolated. Presence of *E. coli* in water is an indication that other pathogens present in feces might also be present.

E. coli causes traveler's diarrhea. Being a versatile opportunistic pathogen, it infects any part of the body subject to fecal contamination. It can cause urinary and reproductive tract infections and infection of abdominal cavity after perforation of the bowel. It can also infect other body parts such as gall bladder, lungs, etc.

Answer (A)

28. Topic: Methods in Microbiology

The cell wall of Gram-negative bacterium is thinner, but more complex than that of a Gram-positive bacterium. Only 10–20% of the cell wall is peptidoglycan, the remainder consists of various polysaccharides, proteins and lipids. That is the reason why, Gram negative bacteria fail to retain the crystal violet dye during the decolorization procedure of Gram staining. These decolorized cells can be rendered visible with a suitable counterstain, which is usually positively charged safranin, which stains them pink.

Answer (A)

29. Topic: Microbial Classification and Diversity (Bacteria)

- *Methanococcus* can tolerate high pressure, so also known as Barophile or piezophile.
- *Dunaliella* also known as halophile can withstand very high concentration of salt.
- *Sulfolobus* also known as acidophile can grow in pH between 2–3.
- *Escherichia* are also known as mesophile, they can grow in moderate temperature.

Answer (D)

30. Topic: Microbial Diseases

Crown gall occurs when a wound on the stem allows the bacteria to invade the plant. *Agrobacterium tumefaciens* is a gram negative rod-shaped, motile bacterial that grows aerobically. It causes crown gall diseases in plants by integrating its own DNA into host genome causing tumors. Crown gall appears as small swellings on the stem or root generally near the soil line (sometimes in aerial parts). Young tumors are soft, spherical, white or cream colored. Older tumors are irregularly-shaped and are brown or black. After infection, the bacteria cause a cancerous proliferation of the stem tissue in the region of the crown. *A. tumifaciens* is now generally used as a tool for introducing desired genes in the plants by genetic engineering.

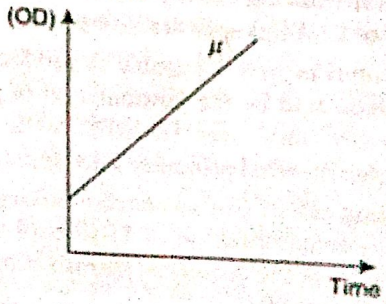
Answer (C)

31. Topic: Microbial Growth and Nutrition

Culture in exponential phase

At $t = 0$, OD at 540 nm (Linearly proportional to the number of cells)

Slope = μ



OD at 2 hrs (OD_2) = 0.3

OD at 4 hrs (OD_4) = 0.6

$$\text{Growth rate} = \frac{\ln(OD_4/OD_2)}{(t_2 - t_1)}$$

$$= \frac{\ln(0.6/0.3)}{(4-2)}$$

$$= \frac{\ln(2)}{(2)} = \frac{0.693}{2}$$

$$= 0.346$$

Answer (0.346)

32. Topic: Viruses—Structure and Classification

Number of cells = 100

Average burst size = 80

Ratio of number of phage particle going for lysogeny to lysis = 4 : 1

Total number of virus cycle (lysogeny + lysis) = 4 + 1 = 5

$$\text{Number of infected cell} = \frac{\text{Total cell}}{\text{Total cell cycle}} = \frac{100}{5} = 20$$

Number of phage particles released = Number of infected cell × Average burst size

$= 20 \times 80 = 1600$

Answer (1600)

33. Topic: Microbial Disease and Host Pathogen Interaction

Neurotoxins are the substances that produce their poisonous effects by acting on the neural system.

- Botulinum toxin (BTX) is a neurotoxin, produced by the anaerobic bacterium *Clostridium botulinum*, found in improperly canned foods, acts at the neuromuscular junction preventing the release of acetylcholine (a chemical released by neuron for muscle cell contraction) and muscle contraction does not occur. It paralyzes in muscles in a relaxed or flaccid state. It starts with small eye muscles, eventually progressing to larynx, pharynx and then respiratory muscles.
- Cholera toxin (CTX) is secreted by the bacteria *Vibrio cholera* and cause gastrointestinal disease. When ingested, *V. cholerae* invades intestinal mucosa where it multiplies and releases enterotoxin that causes nausea, vomiting, abdominal pain and diarrhea.
- Streptolysin-O is an exotoxin produced by *Streptococci* species such as *S. pyogenes*. Streptolysin-O causes a hemolysis (rupturing of red blood cells or erythrocytes) thereby causing cardiovascular disease.
- Diphtheria toxin is secreted by *Corynebacterium diphtheria*, the pathogenic bacterium that causes diphtheria. It is an exotoxin that inhibits protein synthesis and kills susceptible cells.

Answer (C)

34. Topic: Nitrogen Fixation

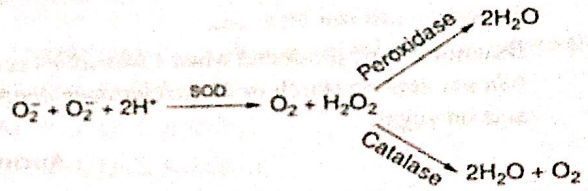
- Tryptophan, a root exudate is converted into Indole acetic acid (plant hormone) by the *Rhizobia* which in turn cause the root hair to grow excessively and curl.
- *Rhizobia* are found in the soil as both free-living organism and as nitrogen fixing symbionts in the root nodules of leguminous plants. Nodules are developed to house nitrogen-fixing bacteria. Most of the genes required for nodulation are carried by *rhizobia*. The compounds such as flavonoids secreted by host plant present root in the rhizosphere reduce *nod* genes in *rhizobia*. These *nod* genes encode several proteins required for synthesis and export of Nod factor by the bacteria. Nod factor is responsible for many developmental changes including root hair deformation and formation of nodule premordium. *Rhizobia* enter from the root hair surface into the root nodule by growing and dividing inside a tubule called an infection thread. Curling of root hairs of host plants takes place under the influence of Nod factors from *rhizobia*.

Answer (D)

35. Topic: Aerobic and Anaerobic Respiration

During metabolism, the reduction of oxygen results in formation of tonic and reactive oxygen species (ROS) such as superoxide onion (O_2^-), hydrogen peroxide (H_2O_2) and hydroxyl radical (OH^-). The aerobic bacteria contain the enzyme superoxide dismutase (SOD) that enable them to overcome the toxic effects of H_2O_2 and O_2^-. All bacteria that can survive in the presence of oxygen contain superoxide dismutase. The obligate anaerobes, on the other hand, lack superoxide dismutase and hence cannot survive when exposed to oxygen as they undergo lethal oblation.

They also lack catalase or peroxidase which is present in other bacteria for further metabolism of H_2O_2.



Answer (B)

36. Topic: Aerobic and Anaerobic Respiration

In anaerobic respiration, usually an inorganic substance etc., (such as nitrate, nitrite sulfate fumarate) is the final electron acceptor as compared to molecular oxygen in aerobic respiration. Oxidation of compounds such as carbohydrates (e.g., glucose), lipids and proteins result in formation of simple organic compounds as well as ATP as the energy source. In aerobic organisms, glucose is oxidized completely as follows:

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$$

In, bacteria, the following pathway of glucose oxidation exists:

a. Glycolytic pathway (most common)

b. Oxidation pentose phosphate pathway (hexose monophosphate shunt)

c. Entner-Doudoroff pathway (in obligate aerobes)

They all lead to formation of pyruvate which further gets decarboxylated to CO_2 via Krebs' cycle in aerobes. The chemical energy is contained in NADH + H^+ and NADPH + H^+ and succinate, that are produced during Krebs' cycle. In the final step of aerobic respiration, that is electron transport oxidation, phosphorylation takes place where ATP is generated.

One molecule of glucose on glycolysis in anaerobic respiration yields 2 molecules of pyruvate and 2 molecules of ATP. In aerobic respiration, 1 molecule of glucose through glycolysis, Krebs' cycle and electron transport chain yields 2 molecules of pyruvate and each molecule of pyruvate is oxidized completely to 3 molecules of CO_2 and 15 molecules of ATP.

Hence, number of ATP molecules generated in aerobic respiration are more than that in anaerobic respiration.

Answer (C)

37. Topic: Applications of Microbiology

Microbes are also used for commercial and industrial production of certain chemicals like organic acids, alcohols and enzymes.

- Examples of acid producers are *Aspergillus niger* (a fungus) of citric acid.
- Cellulase produced by *Trichoderma viride* is used in detergents as it can digest small fibers.
- Vitamin B_{12} (cyanocobalamin) is commonly produced by *Propionibacterium freudenreichii* and *Pseudomonas denitrificans*.
- Butanol can be produced when *Clostridium acetobutylicum* acts on starch or *C. saccharoacetobutylicum* acts on sugar.

Answer (D)

38. Topic: Viruses—Structure and Classification

HIV is classified as a retrovirus since its genetic information is carried in RNA instead of DNA. It consists of an inner core of two single-stranded RNA molecules that are enclosed within the core covered by protein coat (capsid). An envelope composed of a lipid bilayer that is penetrated by glycoproteins surrounds the HIV capsid. A viral enzyme called reverse transcriptase reads the viral RNA strand and makes a DNA copy.

Answer (A)

39. Topic: Microbial Diseases

- Amoebiasis is caused by *Entamoeba histolytica*. *E. histolytica* is a monoxenous, monogenetic parasite completing its life cycle in a single host man and having only the asexual generation in the life cycle. Patients with amoebiasis have abdominal tenderness, 30 or more bowel movements per day and dehydration from excessive fluid loss.
- African sleeping sickness is caused by either of the protozoan *Trypanosoma rhodesiense* or *Trypanosoma gambiense* and spreads through the bite of *tse tse* fly. The initial symptoms are red sore, followed by fever, swollen lymph glands along the back of the neck, shortness of breath, pain in the muscles and joints, headaches and irritability. In the later stage, the parasite enters the CNS causing lethargy and unconsciousness.
- Leishmaniasis or Kala Azar is a chronic disease caused by *Leishmania donovani*, a unicellular parasitic protozoan. It lives intracellularly in the endothelial linings of the blood vessels, spleen, bone marrow, liver and WBCs.
- American sleeping sickness or Chagas' disease is caused by the flagellate protozoan *Trypanosoma cruzi*. It spreads through blood-sucking "kissing bugs". The initial symptoms include mild symptoms such as fever, fatigue, body aches, headache, rash, loss of appetite, diarrhea and vomiting to lethargy and sleepiness when the parasite enters the blood. Then in the later stages the parasite affects the neural and digestive systems as well as heart causing cardiac dilation, damage of digestive tract, etc.

Answer (C)

40. Topic: Antibiotics (Microbial Diseases)

The tetracyclines are a family of antibiotics that inhibit protein synthesis by preventing the attachment of aminoacyl-tRNA to the ribosomal acceptor (A) site. Tetracyclines are broad-spectrum agents, exhibiting activity against a wide range of gram-positive and gram-negative bacteria, atypical organisms such as chlamydiae, mycoplasmas, and rickettsiae, and protozoan parasites.

Answer (D)